SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1
Hour QUESTIONS: 5 Total
Marks:30

Annexure A

Resolution Algorithm
Implementation

Code of Conduct:

I JASMITHA R(192524295)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
JASMITHA R

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground Knowledge
Base

1. If it rains, then the ground becomes wet.
2. It is raining. Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment. Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF. Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member. Goal 1. Prove that **Priya can borrow books** using the Resolution Algorithm. Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test. Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal Tasks

Prove that **the user is granted access** using the Resolution Algorithm.

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

Question 1

Step 1: Propositional Logic Representation

- Let R = "It rains"
- Let W = "The ground becomes wet"
- Premise 1: $R \rightarrow W$ (If it rains, the ground becomes wet)
- Premise 2: R   (It is raining)

Step 2: Goal

Prove: W (The ground is wet)

Step 3: Convert to CNF

```
Clause 1 (from R → W):   ¬R ∨ W
Clause 2 (from R):        R
```

Step 4: Negate the Goal

```
¬W
```

Step 5: Apply Resolution Algorithm

```
Clauses in KB: {¬R ∨ W}, {R}, {¬W}
Step 1: Resolve {R} and {¬R ∨ W}
         → Resolvent: {W}
Step 2: Resolve {W} and {¬W}
         → Resolvent: {}  (Empty Clause, □)
```

Conclusion

Since the Empty Clause (□) is derived, the negated goal ¬W leads to a contradiction. Hence W ("the ground is wet") is PROVED true by resolution.

Question 2

Step 1: Propositional Logic Representation

- Let S = "A student submits the assignment"
- Let M = "The student receives internal marks"
- Premise 1: $S \rightarrow M$ (If a student submits, then receives marks)
- Premise 2: S(Rahul)   i.e. Rahul submitted the assignment

Step 2: Goal

Prove: M(Rahul) — Rahul receives internal marks

Step 3: Convert to CNF

```
Clause 1 (from S → M):   ¬S ∨ M
Clause 2 (from S):        S
```

Step 4: Negate the Goal

```
¬M
```

Step 5: Apply Resolution Algorithm

```
Clauses in KB: {¬S ∨ M}, {S}, {¬M}
Step 1: Resolve {S} and {¬S ∨ M}
         → Resolvent: {M}
Step 2: Resolve {M} and {¬M}
         → Resolvent: {}  (Empty Clause, □)
```

Conclusion

The Empty Clause is derived, so the negation ¬M is contradicted. Hence it is PROVED that Rahul receives internal marks.

Question 3

Step 1: Propositional Logic Representation

- Let L = "A person is a library member"
- Let B = "The person can borrow books"
- Premise 1: L → B (If a member, can borrow books)
- Premise 2: L(Priya) i.e. Priya is a library member

Step 2: Goal

Prove: B(Priya) — Priya can borrow books

Step 3: Convert to CNF

```
Clause 1 (from L → B):   ¬L ∨ B
Clause 2 (from L):        L
```

Step 4: Negate the Goal

```
¬B
```

Step 5: Apply Resolution Algorithm

```
Clauses in KB: {¬L ∨ B}, {L}, {¬B}
Step 1: Resolve {L} and {¬L ∨ B}
```

```
         → Resolvent: {B}
Step 2: Resolve {B} and {¬B}
         → Resolvent: {}  (Empty Clause, □)
```

No further clauses can be generated after the empty clause is reached. **Conclusion**

The Empty Clause is derived, refuting ¬B. Hence it is PROVED that Priya can borrow books.

Question 4

Step 1: Propositional Logic Representation

- Let A = "A student clears the aptitude test"
- Let E = "The student is eligible for placement"
- Premise 1: A → E (If cleared aptitude test, then eligible)
- Premise 2: A(Arun) i.e. Arun cleared the aptitude test

Step 2: Goal

Prove: E(Arun) — Arun is eligible for placement

Step 3: Convert to CNF

```
Clause 1 (from A → E):   ¬A ∨ E
Clause 2 (from A):        A
```

Step 4: Negate the Goal

```
¬E
```

Step 5: Apply Resolution Algorithm

```
Clauses in KB: {¬A ∨ E}, {A}, {¬E}
Step 1: Resolve {A} and {¬A ∨ E}
         → Resolvent: {E}
Step 2: Resolve {E} and {¬E}
         → Resolvent: {}  (Empty Clause, □)


Resolution Tree:
   {A}        {¬A ∨ E}
      \         /
         {E}           {¬E}
            \           /
               {}  (□)
```

Conclusion

The resolution tree terminates in the Empty Clause, so ¬E is contradicted. Hence it is PROVED that Arun is eligible for placement.

Question 5

Step 1: Propositional Logic Representation

- Let P = "The user enters the correct password"
- Let A = "The user is authenticated"
- Let G = "The user is granted access"
- Premise 1: P → A   (Correct password ⇒ authenticated)
- Premise 2: A → G   (Authenticated ⇒ granted access)
- Premise 3: P   (The user entered the correct password)

Step 2: Goal

Prove: G — The user is granted access

Step 3: Convert to CNF

```
Clause 1 (from P → A):  ¬P ∨ A
Clause 2 (from A → G):  ¬A ∨ G
Clause 3 (from P):      P
```

Step 4: Negate the Goal

```
¬G
```

Step 5: Apply Resolution Algorithm

```
Clauses in KB: {¬P ∨ A}, {¬A ∨ G}, {P}, {¬G}
Step 1: Resolve {P} and {¬P ∨ A}
        → Resolvent: {A}
Step 2: Resolve {A} and {¬A ∨ G}
        → Resolvent: {G}
Step 3: Resolve {G} and {¬G}
        → Resolvent: {}  (Empty Clause, □)
```

Conclusion

Each resolution step eliminates one complementary literal pair (P with ¬P∨ A gives A; A with ¬A∨ G gives G; G with ¬G gives □). Since the Empty Clause is reached, ¬G is contradicted. Hence it is PROVED that the user is granted access.